DO NOW — RECALL

Lesson 129 · Dictionary Compression and Repeated Subs

Name: _____ Class: _____ Date: _____

1. RLE is effective on what kind of input?

2. How many bits are in three bytes?

3. Which memory is typically volatile: RAM or ROM?

4. What is the binary result of 1 + 1 in a full adder with carry-in 0?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Long consecutive runs of identical symbols.

2. Model answer

24.

3. Model answer

RAM.

4. Model answer

Sum 0, carry-out 1.

By the end, you can…

• Locate useful repeated words or substrings.

• Assign tokens to recurring sequences and store mapping.

• Replace sequences with tokens; decode using the same mapping.

Vocabulary — know these terms

dictionary coding — Encodes recurring sequences by referring to dictionary entries.

token — Short code representing a longer sequence.

overlap — Repeated sequence used by more than one part of the source.

lossless — Reconstruction is exact when dictionary and token mapping are known.

NEW LEARNING

Read, annotate and explain

Find recurring text

Locate useful repeated words or substrings.

Build dictionary

Assign tokens to recurring sequences and store mapping.

Encode and decode

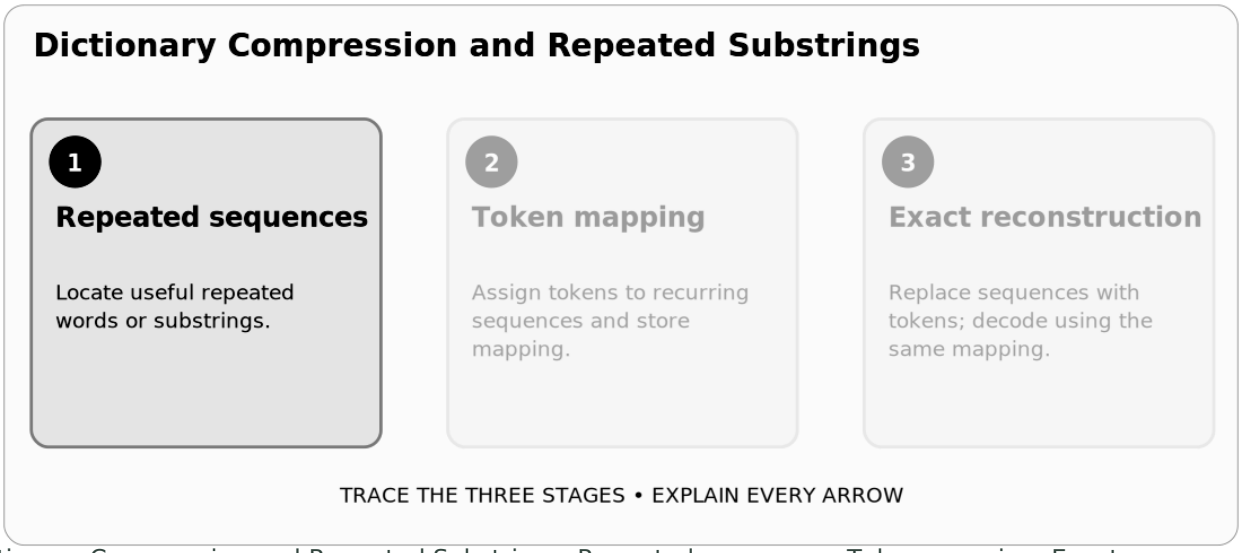
Replace sequences with tokens; decode using the same mapping.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

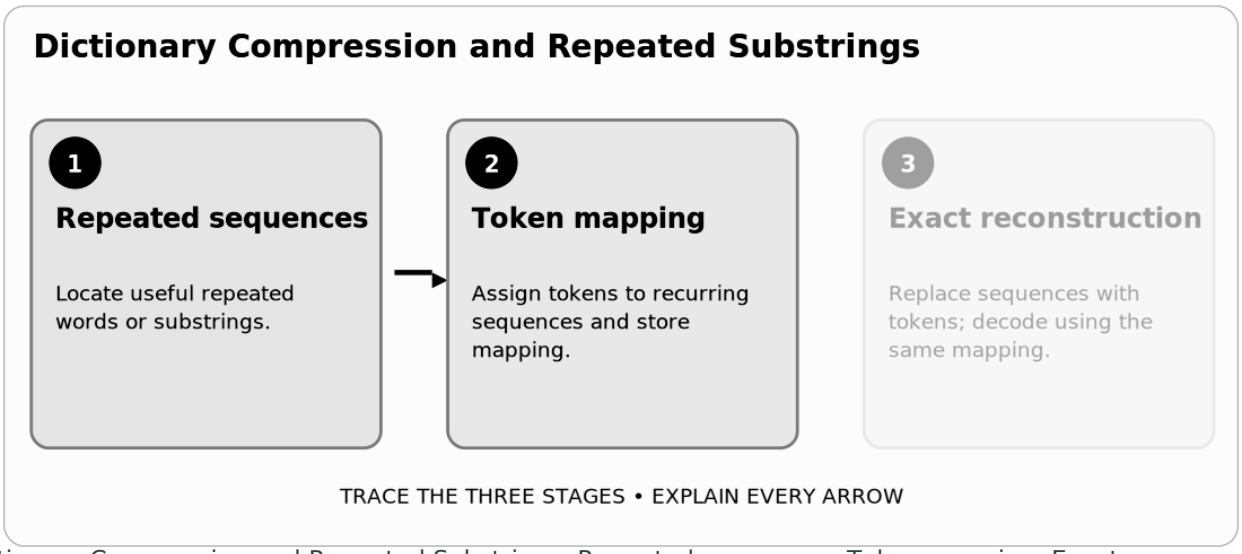
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Dictionary Compression and Repeated Substrings: Repeated sequences; Token mapping; Exact reconstruction

New Learning

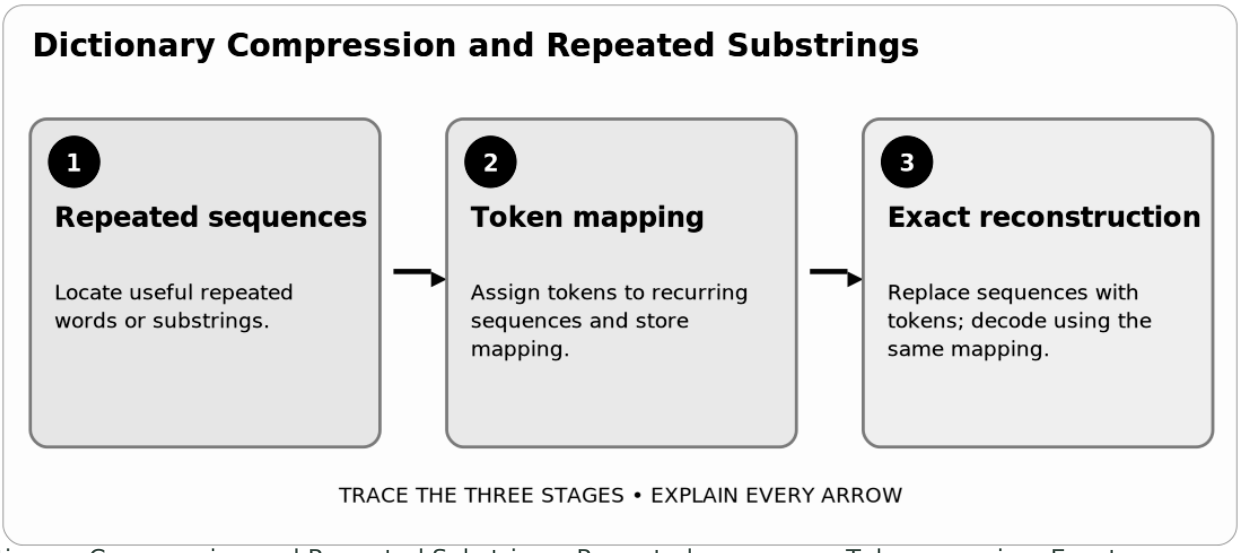


Dictionary Compression and Repeated Substrings: Repeated sequences; Token mapping; Exact reconstruction

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Dictionary Compression and Repeated Substrings: Repeated sequences; Token mapping; Exact reconstruction

I DO — WORKED EXAMPLE

Follow the worked method

Use dictionary {#1:'COMPUTER', #2:'SCIENCE'} to encode 'COMPUTER SCIENCE COMPUTER'.

Two dictionary entries

Predict the first step from the prompt.

#1 #2 #1

Explain the intermediate result or decision.

Restore original phrase

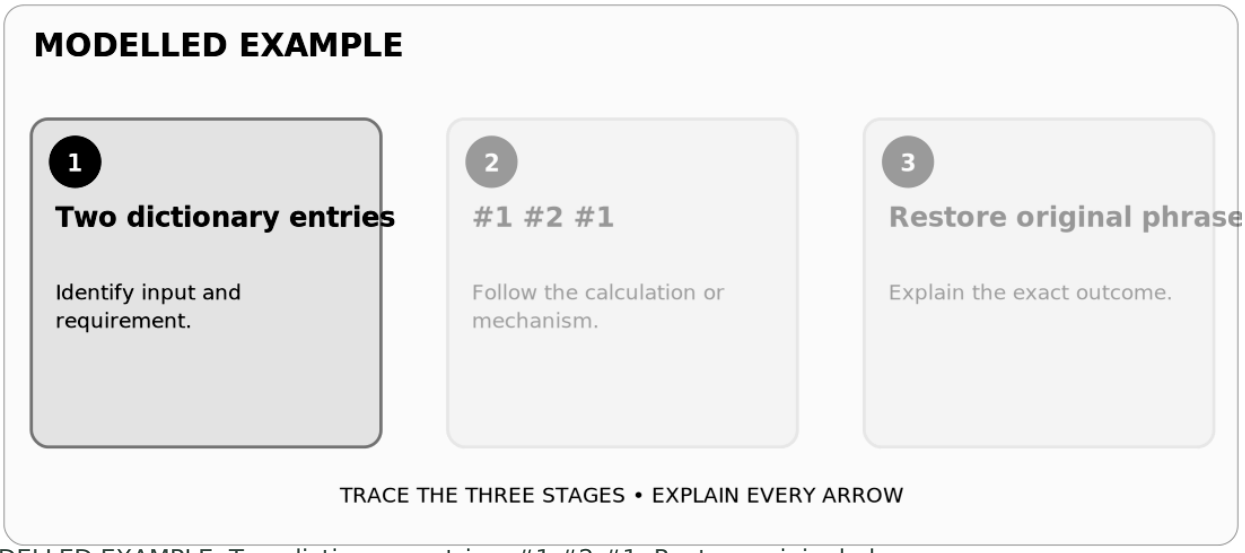
Treat spaces as retained separators: #1 #2 #1. Decoder substitutes entries to recover 'COMPUTER SCIENCE COMPUTER'. Include dictionary storage when calculating total size.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Use dictionary {#1:'COMPUTER', #2:'SCIENCE'} to encode 'COMPUTER SCIENCE COMPUTER'.

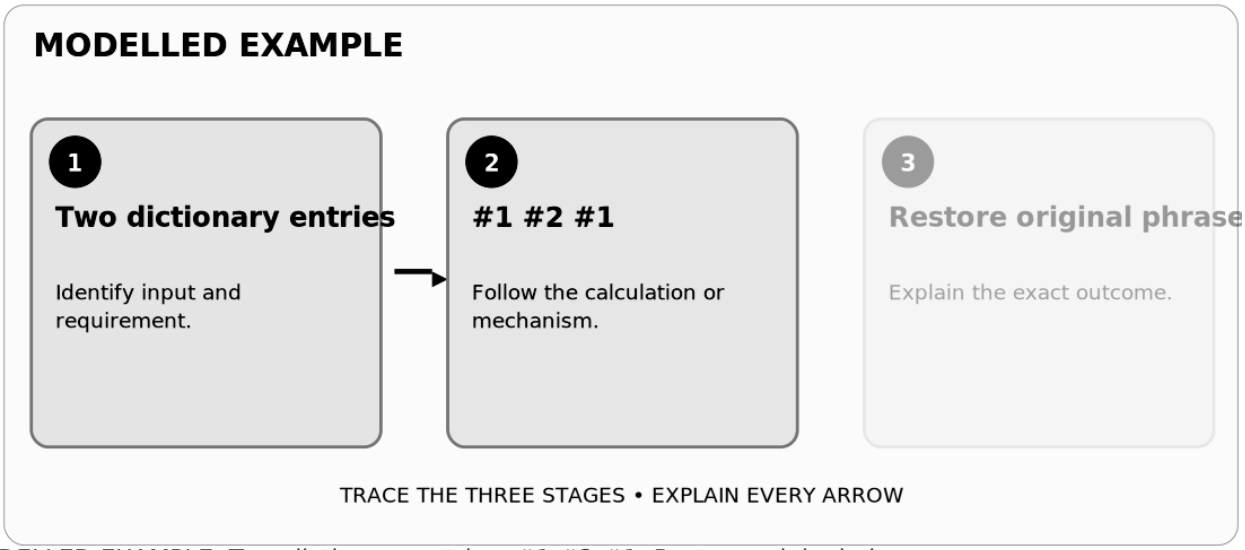
Predict the first step from the prompt.



MODELLED EXAMPLE: Two dictionary entries; #1 #2 #1; Restore original phrase

Use dictionary {#1:'COMPUTER', #2:'SCIENCE'} to encode 'COMPUTER SCIENCE COMPUTER'.

Predict the first step from the prompt.



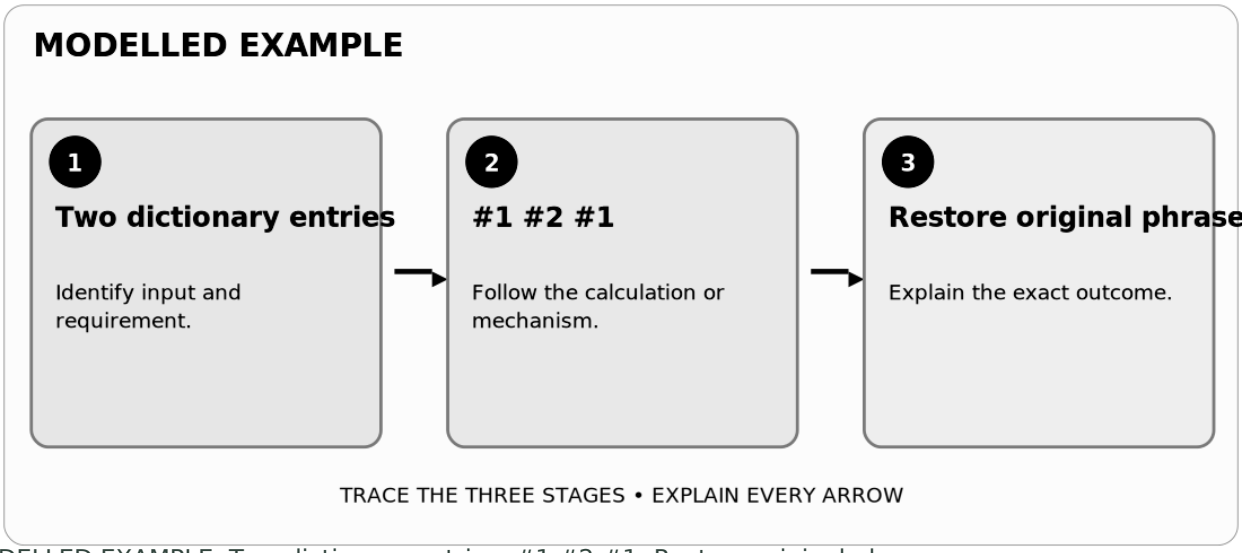
MODELLED EXAMPLE: Two dictionary entries; #1 #2 #1; Restore original phrase

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Use dictionary {#1:'COMPUTER', #2:'SCIENCE'} to encode 'COMPUTER SCIENCE COMPUTER'.

Predict the first step from the prompt.



MODELLED EXAMPLE: Two dictionary entries; #1 #2 #1; Restore original phrase

WE DO — GUIDED PRACTICE

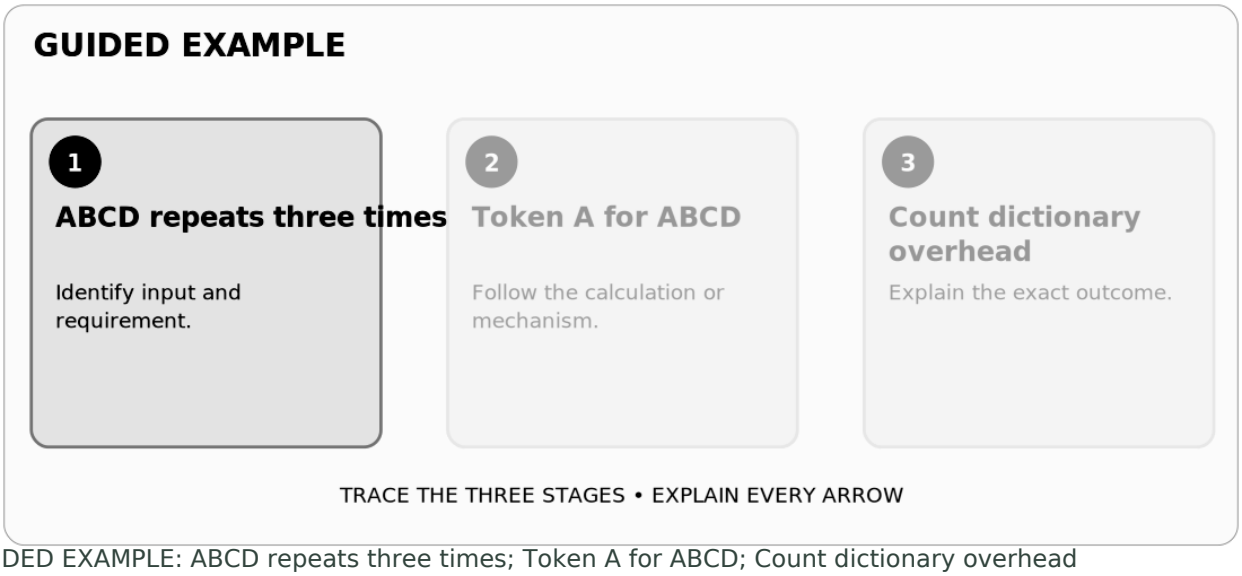
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Dictionary {A:'ABCD'}; encode 'ABCDABCDXYZABCD' and discuss the result.

Predict the first step from the prompt.



GUIDED EXAMPLE: ABCD repeats three times; Token A for ABCD; Count dictionary overhead

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: ABCD repeats three times Predict the first step from the prompt.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain dictionary compression and why the decoder needs the dictionary. [6]

Question 2

Using entries {1:'RED',2:'BLUE'}, encode 'RED BLUE RED RED' and decode '2 1'. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate dictionary coding for a document with frequent terms versus already-compressed random-looking data. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Dictionary coding requires permanently discarding repeated words.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

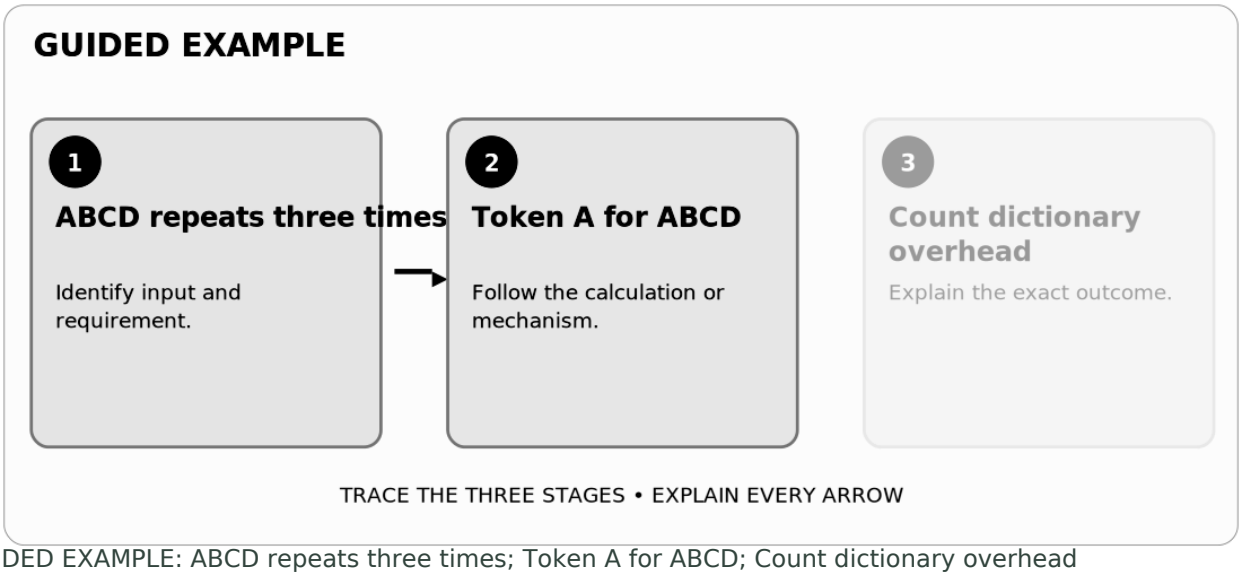
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Dictionary {A:'ABCD'}; encode 'ABCDABCDXYZABCD' and discuss the result.

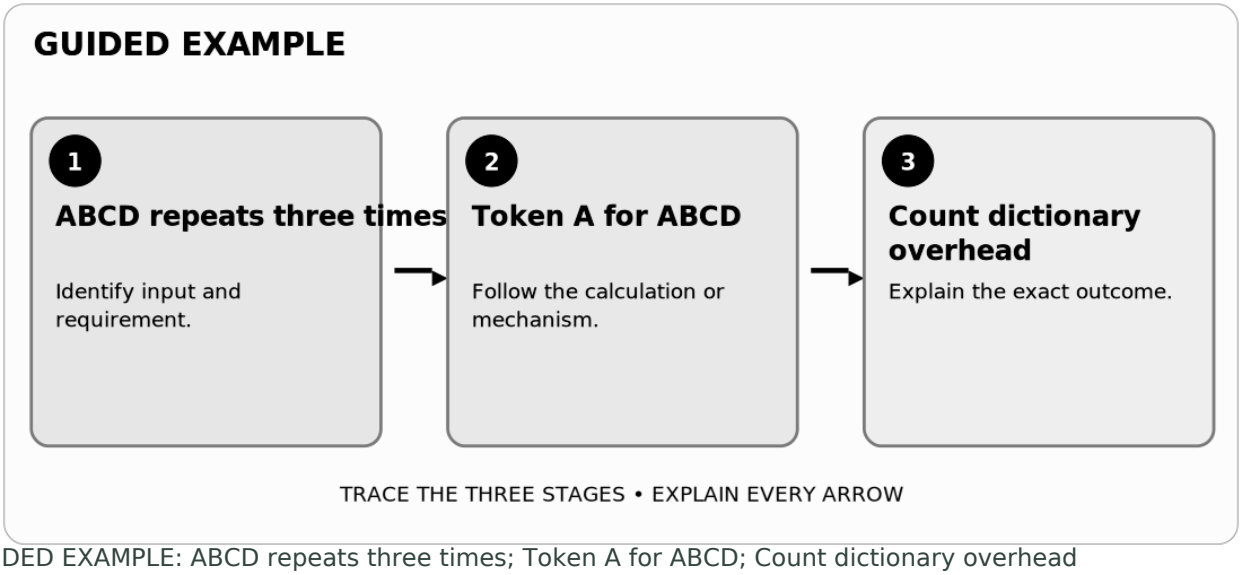
Predict the first step from the prompt.



GUIDED EXAMPLE: ABCD repeats three times; Token A for ABCD; Count dictionary overhead

Dictionary {A:'ABCD'}; encode 'ABCDABCDXYZABCD' and discuss the result.

Predict the first step from the prompt.



GUIDED EXAMPLE: ABCD repeats three times; Token A for ABCD; Count dictionary overhead

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Token A for ABCD

Explain the intermediate result or decision.

Count dictionary overhead

AAXYZA, using single-character token A and leaving XYZ as literals under this toy scheme. Original 15 characters become 6 token/literal characters, plus dictionary overhead; real formats require unambiguous token conventions.

IMPROVE: Explain one correction to your own reasoning.